

Iris Machine Learning Model - Summary Report

Objective

Build a classification model using the Iris dataset to predict flower species.

Dataset

The Iris dataset from scikit-learn contains 150 samples, 4 numerical features, and 3 target classes: Setosa, Versicolor, and Virginica.

Exploratory Data Analysis

The dataset was inspected for structure, summary statistics, missing values, and class distribution. No missing values were found.

Feature Selection

Features: sepal length, sepal width, petal length, petal width. Target: species.

Model

The dataset was split into 80% training and 20% testing data. A Logistic Regression classifier was trained.

Evaluation

The model was evaluated using Accuracy, Classification Report (Precision, Recall, F1-score), and Confusion Matrix. The model achieved excellent performance (approximately 100% accuracy).

Conclusion

Logistic Regression successfully classified the Iris flower species with high accuracy, demonstrating its effectiveness for this dataset.